Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years shipping **React**, **Next.js**, and TypeScript web apps with Python and Node.js APIs, SQL and NoSQL databases, **LLM**-powered features, and CI/CD deployments on AWS and Vercel.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized PostgreSQL schemas and **FastAPI** services for a multi-tenant AI education platform serving 500+ active users, tuning indexed queries to cut dashboard latency from 1.2s to 350ms.
- Restructured fault-tolerant **LLM** job processing with background workers and failure isolation, handling 1K+ AI jobs at 3x throughput, and shipped integrations through CI/CD-backed releases at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a full-stack product end to end with a **React** frontend, Flask REST APIs, and MySQL on AWS, deploying via GitHub Actions to Vercel for 150+ instructors and 10K+ daily requests.
- Built **AI-powered features** with GPT-4o, Claude, and Qwen-VL that generate rubrics and feedback from uploaded content, with fallbacks and per-call cost tracking, cutting instructor feedback turnaround 60% in production.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Cut API payloads for **low-bandwidth** users with pagination, field filtering, and caching of frequently requested data, improving response times 30% across large-scale integrations used by distributed clients.
- Wrote 20+ automated unit and integration test suites run nightly in Jenkins CI and redesigned AWS Glue ETL workflows, boosting pipeline efficiency 68% and **reducing technical debt** 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a TypeScript React and Node.js platform with JWT authentication, role-based access, and REST APIs over MongoDB, syncing 2M+ records across multiple sites with **real-time** WebSocket updates for every site.
- Used **Google Analytics** to study feature usage, load times, and drop-off points, guiding fixes that cut resource wastage 42% and raised lead conversions 15% through Jenkins CI/CD releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a **mobile-first** React Native and React TypeScript experience with Flask APIs and Firebase real-time updates, keeping responses under 200ms and improving cross-platform compatibility 15% across Android, iOS, and web.
- Built pricing and coupon logic that recomputed totals with eligibility and expiry rules, and hourly Spark analytics on active users and **usage trends** for internal dashboards across 50K+ records.

Skills

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS, Java
Frontend: React, Next.js, React Native, TailwindCSS, Shadcn, Responsive Design
Backend: Node.js, Express.js, FastAPI, Flask, Django, REST APIs, WebSockets, Selenium
Data & AI: PostgreSQL, MySQL, MongoDB, Firebase, LLM APIs, Prompt Engineering, Embeddings, Semantic Search
Cloud & Practices: AWS, Vercel, Docker, GitHub Actions, CI/CD, Git, Code Reviews, Jest, PyTest, Google Analytics, Agile

Projects

Interactive Recipe Platform | *Next.js, TypeScript, PostgreSQL, Prisma, Vercel*

- Designed a **Next.js** recipe site with serving-size scaling, unit conversion, step-by-step cook mode, shopping lists, recipe **schema markup**, optimized images, and Core Web Vitals tracking through GA4.

AI Ingredient Substitution Assistant | *Python, FastAPI, LLM APIs, Embeddings, PostgreSQL*

- Designed an **AI** feature that suggests ingredient substitutions and similar recipes using embedding search over a recipe library, with dietary filters and an evaluation set checking **suggestion quality**.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex, Shadcn*

- Built an AI content platform with Clerk sign-in, FAISS and Pinecone **semantic search**, summarization, and real-time collaboration over 50K+ documents, cutting document review time 60% for 200+ users.